Quiz 3

Graded

Student

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Total Points

100 / 100 pts

Question 1

Pre-trained LM

20 / 20 pts

✔ **+ 20 pts** Correct

+ 0 pts Incorrect

Question 2

Reasoning

20 / 20 pts

✔ **+ 20 pts** Correct

+ 0 pts Incorrect

Question 3

Zero-shot vs Few-shot reasoning

20 / 20 pts

✔ **+ 20 pts** Correct

+ 0 pts Incorrect

Question 4

Pre-training

20 / 20 pts

+ 0 pts Incorrect

✔ **+ 20 pts** Correct

Question 5

Self-Supervised Learning

20 / 20 pts

+ 20 pts Correct

✔ **+ 20 pts** Correct

Q1 Pre-trained LM

20 Points

Which of the following statements is NOT TRUE when comparing GPT, BART, and BERT?

- ( ) All three models can be implemented using the Transformer architecture.
- (•) BART cannot perform tasks such as sentence classification, which are typical applications for BERT.
- ( ) BART uses a sequence-to-sequence Transformer architecture, while BERT uses an encoder-only Transformer architecture.
- ( ) GPT is a decoder-only model designed for language modeling.

Q2 Reasoning

20 Points

In the context of Large Language Models (LLMs), which of the following is a core characteristic of 'reasoning'?

- ( ) The capability to memorize and retrieve exact facts from a training corpus without modification.
- (•) The ability to produce answers by combining information through multiple intermediate steps.
- ( ) Reducing the number of hidden layers in a neural network to speed up inference.
- ( ) Increasing the size of the training dataset to improve the performance.

Q3 Zero-shot vs Few-shot reasoning
20 Points

In prompt-based reasoning, what distinguishes a 'few-shot' approach from a 'zero-shot' approach?

- (●) Few-shot provides examples in the prompt, while zero-shot uses only a instructional phrase.
- ( ) Few-shot prompts are limited to a single step, while zero-shot prompts allow for multi-step chains.
- ( ) Few-shot requires updating model weights, whereas zero-shot is strictly for inference.
- ( ) Zero-shot is usually more accurate for complex math problems compared to few-shot prompts.

Q4 Pre-training
20 Points

Why is 'Pre-training' considered a way to unlock the power of unannotated data?

- (●) It uses self-supervised learning to learn general representations from massive amounts of raw text.
- ( ) It reduces the model size so it can be run on smaller consumer devices.
- ( ) It eliminates the need for any fine-tuning or task-specific training later on.
- ( ) It converts unannotated text into human-readable parse trees automatically.

Q5 Self-Supervised Learning
20 Points

Provide one source of text data that a large language model (LLM) can use for self-supervised training.

Books